Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI, Agentic Workflows, Prompt Orchestration, Retrieval-Augmented Generation (RAG), Large Language Models, Autonomous Anomaly Detection, Supervised Machine Learning, Time-Series Modeling (SARIMA), MLOps, Distributed Data Engineering, Dimensional Modeling (Star/Snowflake), Data Governance & Lineage.

Programming Languages & Tools: Python, PyTorch, TensorFlow, Scikit-Learn, Hugging Face, KIRO IDE, Azure Databricks, Azure Data Factory, Azure Synapse, AWS S3, Redshift, Snowflake, Apache Spark, PySpark, Power BI (DAX), Tableau, Collibra, Docker, Kubernetes, Git.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior AI & Data Platforms Engineer — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (KIRO IDE, prompt orchestration, Copilot) to automate complex schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered and operated autonomous machine learning anomaly detection** frameworks identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream data integrity and compliance reporting.
- **Developed scalable predictive machine learning models** (XGBoost, Random Forests, Scikit-Learn) to predict customer behavior and operational risks, delivering actionable insights that improved enterprise service reliability by 20%.
- **Built automated evaluation and testing frameworks for conversational AI** (SpectrumIQ enterprise virtual assistant), cutting manual QA turnaround by 50% and accelerating secure, production-grade release cycles.
- **Designed high-throughput distributed cloud data pipelines** and dimensional models using Azure (ADF, Databricks, Synapse, SQL) and AWS (S3, Redshift) to process hundreds of terabytes under strict 99.9% availability SLAs.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Architected advanced time-series forecasting models** (SARIMA) to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered automated source-to-target reconciliation firewalls** and governance lineage models using Collibra, reducing operational audit turnaround times by over 80%.
- **Translated complex transactional telemetry into executive decision intelligence** using advanced Power BI (DAX) and Tableau semantic models to drive corporate risk management and operational strategy.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed enterprise Key Operating Metric dashboards** in Power BI and Tableau, integrating multi-source telemetry to empower stakeholders with real-time risk visibility and performance tracking.
- **Implemented rigorous data quality validation pipelines** to support large-scale financial and operational reporting frameworks with uncompromised accuracy.
- **Streamlined data governance protocols** by establishing automated lineage mapping and reconciliation mechanisms to meet strict corporate compliance standards.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.